INCH-POUND
MIL-P-24441/41(SH)
23 July 1991

MILITARY SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, BLACK NONABRASIVE NONSLIP, FORMULA 163, TYPE IV

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-P-24441.

FORMULA. This formula covers black nonslip epoxy-polyamide paint designated Navy Formula 163, type IV, for exterior use that is nonabrasive to aircraft arrester wires. The paint shall consist of the ingredients specified in the quantities specified.

Component A

Ingredients	Pounds
Thixotrope 1/	-----
Polyamide 2/	10.6
Polyamide adduct 3/	151.4
Magnesium silicate 4/	220.5
Carbon black 5/	17.2
Titanium dioxide 6/	12.6
Aluminum grit 7/	756.3
Naphtha 8/	149.6

Component B

Ingredients	Pounds
Thixotrope 1/	-----
Epoxy resin 9/	281.2
Magnesium silicate 4/	135.4
Aluminum grit 7/	756.3
Naphtha 8/	123.0

See footnotes on next page.

AMSC N/A FSC 8010
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. Components A and B shall be mixed 1:1 by volume for mixed component tests. Tests shall be performed in accordance with the general specification.

Requirements	Component A		Component B		Mixed components	
	Min	Max	Min	Max	Min	Max
Pigment content, percent	85.8	89.9	88.6	92.6	---	---
Volatiles, percent	9.3	11.3	7.4	9.4	---	---
Nonvolatile vehicle, percent	10.2	14.2	19.6	23.6	---	---
Water, percent	---	1.5	---	0.5	---	---
Pounds per gallon (lb/gal)	12.5	13.5	12.3	13.3	12.4	13.4
Dry-hard, hours at 73°F	---	---	---	---	---	6
Flash point, °F	99	---	100	---	99	---
Titanium dioxide, percent of pigment	1.7	---	---	---	---	---
Pot life, hours at 73°F	---	---	---	---	6	---
Sag resistance, mils	---	---	---	---	15	---
Weight per epoxy, vehicle	---	---	175	195	---	---
Contrast ratio, 3 mil dry film	---	---	---	---	0.98	---
Epichlorohydrin content (vehicle), parts per million	---	---	---	2	---	---
VOC, ASTM D 3960, lb/gal 10/	---	---	---	---	---	2.8

1/ Thixotrope to be used is the manufacturer's choice. In the development of the components A and B, 8.8 pounds (1.0 gallons) of Thixatrol ST made by NL Industries were used in each component. Manufacturer is responsible for choosing a thixotrope that meets all the requirements herein, including shelf life. Thixotrope is a pigment for calculation purposes.

2/ GENAMID 2000, General Mills Chemicals; UNIREZ 2810, Union Camp; AZAMIDE 600, AZ Products; ANCAMINE 507, Pacific Anchor Chemical Corp.; EPOTUF SF7791, Reichhold Chemical.

3/ VERSAMID 280B75, General Mills Chemicals; UNIREZ 2180B73, Union Camp; AZAMIDE 680B75, AZ Products; ANCAMINE 700B75, Pacific Anchor Chemical Corp.; EPOTUF SF7792, Reichhold Chemical.

4/ Mistron 400, "Mistron 500", Cyprus Industries.

5/ TT-P-350

6/ Titanium dioxide conforming to ASTM D 476, type IV. In the development of component A, DuPont Tipure R960 was used.

7/ LSA-284, Reynolds.

8/ Conforming to ASTM D 3734, type I. In the development of components A and B, AMSCO Super High Flash Naphtha was used.

9/ EPON 828, Shell Chemical Co.; ARALDITE 6010, Ciba-Geigy Corp.; DER 331, Dow Chemical Corp.; AZEPOXY 128, AZ Products; EPO-TUF 37-140, Reichhold Chemical Corp.; EPI-REZ 510, Celanese.

10/ For VOC calculations, component A makes approximately 102 gallons and component B makes approximately 102 gallons.

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Use of alternate ingredients in this formula must have prior approval of the Naval Sea Systems Command. Approval will be based on review of data demonstrating equivalent physical, chemical and performance characteristics of paint manufactured with the proposed alternate material. Development of the required data package is the responsibility of the proposing agent. Paint incorporating the proposed alternate ingredient shall conform to all the requirements of this military specification sheet and the general specification.

Preparing activity:
Navy - SH
(Project 8010-N260)

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